Does PIX Usage Move the Fees of Cards? A Quantitative Check on Brazilian Credit Card Interest Rates and Merchant Fees

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Does PIX Usage Move the Fees of Cards? A Quantitative Check on Brazilian Credit Card Interest Rates and Merchant Fees

PIX is widely credited with lowering the cost of retail payments in Brazil. This paper checks a narrower version of that claim: whether how often PIX is used tracks what Brazilians actually pay for cards. Two costs matter here. One is interest on unpaid credit card debt, which comes in two forms: a "revolving" rate that applies the moment a bill goes unpaid, and an "installment" rate, a calmer fixed-payment plan that debt is required by law to convert into after one billing cycle. The other is the merchant fee, what a business pays to accept a card payment, separate from anything the customer pays. No published study has tested either against PIX usage directly. Using the Central Bank of Brazil's own interest-rate data from 2019 to 2026, alongside reconstructed PIX transaction-volume figures, this paper finds no evidence that PIX's growth has pushed card costs down. If anything, the opposite is true: the revolving rate climbed from 304% to 452% during exactly the years PIX usage exploded, moving in step with it ($r = 0.92$). The installment rate barely moved (174% to 186%) and tracks PIX far more loosely ($r = 0.67$). Both point away from a competition story and toward Brazil's interest-rate cycle, and a card market that runs on its own regulatory logic. Merchant fees tell a similar story: debit fees have been capped by law since 2018, two years before PIX existed, and unregulated credit card fees show no sign of moving toward PIX's price even after five years of PIX eating into card volume. Despite the correlation being real and strong, five years of data cannot turn it into proof of cause and effect.

I. Introduction and Research Question

Compared to the world average Brazil's credit card market is extremely large and expensive. On cost, merchants pay roughly 2.2%–5% of their total transactional amount to accept a credit card, against about 1.7% in the United States, 1.5% in Canada, and 0.3% in the European Union.[1] On size, Brazil runs the second-largest real-time payments market in the world, processing tens of billions of PIX transactions a year.[2][3] Before PIX, cash and cards handled nearly all retail payments in the country. PIX arrived as a free alternative for individuals and a cheap one for merchants, and it grew fast, so that by 2023 it was already outpacing credit and debit cards combined in transaction count.[4] The obvious next question is whether that growth actually put any pressure on card prices.

This paper checks whether PIX usage tracks what people pay in card interest, and what merchants pay in card fees, which the existing research on PIX has not directly tested. Three studies come closest, and each looks at something different. Sarkisyan looks at bank deposits, and finds PIX gave small banks a real way to compete with big ones for savings.[5] Xu looks at lending across the fintech sector broadly, with PIX as one part of that, and finds rates fell as competition increased.[6] Sampaio and Ornelas look at how many new bank accounts and credit relationships PIX helped create.[7] None of the three measures card interest or merchant fees. If PIX's competitive effect reaches into card pricing at all, it should show up as a link between PIX's growth and a fall in one of these two costs, which is exactly what this paper checks, using the Central Bank of Brazil's own published data.

II. Existing Research

Two things in the existing research come close to this question, and both are one-time snapshots rather than trends: a snapshot is a single measurement, for example "PIX costs merchants 0.22% today," while a trend is that same measurement tracked repeatedly to see if it moves. Duarte,

[1] Duarte, Angelo, et al. "Central Banks, the Monetary System and Public Payment Infrastructures: Lessons from Brazil's Pix." BIS Bulletin no. 52, Bank for International Settlements, Mar. 2022.

[2] "What Is Pix and How Is It Driving Brazilian E-Commerce?" *Volt*, 2022.

[3] EBANX. "Five Years On, Pix Approaches 8 Monthly Transactions." *EBANX*, Nov. 2025.

[4] Matera. "Pix by the Numbers." *Matera*, 2024–2025.

[5] Sarkisyan, Sergey. "Instant Payment Systems and Competition for Deposits." SSRN Working Paper no. 4176990, 2024.

[6] Xu, Rui. "Fintech Competition and Banks' Shrinking Margins in Brazil." IMF Working Paper WP/26/7, International Monetary Fund, 2026.

[7] Sampaio, Matheus C., and José Renato H. Ornelas. "Payment Technology Complementarities and Their Consequences on the Banking Sector: Evidence from Brazil's Pix." BIS Papers no. 152, Bank for International Settlements, 2024.

Frost, Gambacorta, Koo Wilkens, and Shin, writing for the Bank for International Settlements, put PIX's merchant cost at roughly 0.22% per transaction against 2.2% for credit cards.[8] Worth clarifying what "PIX costs merchants" means, since PIX itself is free for the person paying or getting paid: a business still pays its bank a small fee to receive that payment, just a fraction of what card networks charge. Either way, that 0.22% versus 2.2% comparison is a single measurement at one point in time. Separately, current sources put Brazilian credit card fees at 2%–5% as of 2025–2026,[9] roughly the same range cited before PIX existed.

No study builds a multi-year series of card rates or fees and checks it against PIX volume; that evidence simply does not exist yet. The studies above were built to answer different questions, with data suited to those questions, not to card pricing and fees. What follows is a first attempt to close that gap, not a substitute for the kind of dedicated study this question deserves.

III. Data and Methodology

Two interest-rate series come directly from the Central Bank of Brazil's public data. The revolving rate (SGS series 22022) is what a cardholder pays once an unpaid balance rolls over into open-ended debt. It is a national average across every card issuer in Brazil, not any one bank's number, reported already annualized, and available monthly from January 2019 through mid-2026.[10]

The installment rate (SGS série 25478) is what a cardholder pays once that revolving balance gets converted into a fixed-payment plan, something Brazilian law requires after one billing cycle, roughly the month between one statement and the next.[11] It is also a national average, but the Central Bank reports it monthly rather than annualized, so it is compounded here into an annual figure for comparison with the revolving rate.

PIX usage is the least documented variable. No official monthly transaction count was available through the tools used for this research, so the figures below are annual averages re-

[8] Duarte, Angelo, et al. "Central Banks, the Monetary System and Public Payment Infrastructures: Lessons from Brazil's Pix." BIS Bulletin no. 52, Bank for International Settlements, Mar. 2022.

[9] "Credit Cards in Brazil 2026: Rates, the Rotativo Cap, IOF." *globecreditcards.com*, July 2026.

[10] Banco Central do Brasil. "Taxa Média de Juros das Operações de Crédito com Recursos Livres — Pessoas Físicas — Cartão de Crédito Rotativo." SGS Série 22022, Banco Central do Brasil, api.bcb.gov.br.

[11] Banco Central do Brasil. "Taxa Média de Juros das Operações de Crédito com Recursos Livres — Pessoas Físicas — Cartão de Crédito Parcelado." SGS Série 25478, Banco Central do Brasil, api.bcb.gov.br.

constructed from the Central Bank's own reporting,[12] supplemented by industry estimates from Matera[13] and EBANX.[14] The 2023 and 2024 numbers are real annual totals divided by twelve; 2021 and 2022 are rougher, order-of-magnitude estimates, since no precise count for PIX's first two years can be found.

Selic, Brazil's benchmark policy rate and roughly its version of the US federal funds rate, is included as the obvious rival explanation for any movement in card rates that isn't PIX.

Since none of these series break down by bank, everything below describes the market as a whole. With only five annual points, the method stays simple: a Pearson correlation, and for the revolving rate, a partial correlation controlling for Selic. Five points are not enough for a real regression, so nothing here claims statistical significance.

IV. Results

Revolving credit card rate

Year	Revolving rate (%)	PIX volume (B txns/mo.)	Selic (%)
2021	335.6	0.50 (interpolated)	5.9
2022	380.6	1.50 (interpolated)	12.9
2023	438.0	3.50 (actual)	12.9
2024	430.1	5.33 (actual)	10.9
2025	452.3	6.30 (actual)	14.9

The revolving rate correlates with PIX volume at $r = 0.92$, and with Selic at $r = 0.82$. Controlling for Selic, the correlation with PIX volume only drops to 0.88. Put in plain terms, the rate averaged 304% in 2019, the last year before PIX existed, and 452% in 2025, a 148-point jump across exactly the years PIX usage went from nothing to several billion transactions a month.

Installment credit card rate

[12] Banco Central do Brasil. "Relatório de Política Monetária." Banco Central do Brasil, Dec. 2025.
[13] Matera. "Pix by the Numbers." *Matera*, 2024–2025.
[14] EBANX. "Five Years On, Pix Approaches 8 Monthly Transactions." *EBANX*, Nov. 2025.

Year	Installment rate (annualized, %)
2019	174.2
2020	150.7
2021	166.2
2022	178.4
2023	194.5
2024	183.2
2025	186.3

This one correlates with PIX volume at just 0.67, and moved only 12 points over the same stretch, from 174% to 186%. Its 2020 dip lines up neatly with the pandemic-era Selic cut to 2%, the lowest point in the whole series.

V. Interpretation of Findings

Given how little evidence there is to work with, the only relationship the data actually shows is a strong, same-direction correlation between PIX volume and the revolving rate ($r = 0.92$). This correlation runs opposite to what a competition effect would predict: if PIX were putting real price pressure on cards, PIX volume and card rates should move apart, with more PIX trsactional volume meaning lower rates. Instead they rose together. That alone rules out the specific claim this paper set out to test.

Even that correlation should not be trusted too far. A strong correlation is not proof of a cause. A relationship this strong, over that short a stretch, is more likely two unrelated trends that both happened to climb across the same years. Controlling for Selic barely moves the number, from 0.92 down to 0.88, which does not mean Selic fails to explain it; it just means five points cannot tell the difference between “Selic explains it,” “some other shared trend explains it,” and “there is nothing real here at all.”

There is also a structural reason the revolving rate was never likely to respond to PIX. It functions as a penalty for missing a payment, not a price banks compete on for customers, the way a deposit rate is. Since January 2024, a law caps how large that debt can grow: interest, fines, and fees combined cannot push it past double the original amount, so R$1,000 owed tops out at R$2,000 total. The monthly rate itself is untouched by that law, though, and it has stayed flat to rising since the cap took effect.[15] The law stopped the debt from spiraling past double the original amount, but left the underlying intrest rate untouched.

The installment rate makes a useful check on that reading. It behaves more like an ordinary loan than the revolving rate does, so if PIX were pushing down card credit broadly, this is where it should show up first. Instead it correlates more weakly with PIX (0.67 against 0.92) and barely moved at all, which fits the reasoning that the rate is driven by Selic and general credit conditions rather than PIX. Neither product behaves as if it were competing with PIX.

VI. Merchant Fees

Merchant fees work differently from interest rates because a large piece of the picture is set by regulation, not competition, and that regulation predates any possible PIX effect.

Debit interchange has been capped since October 2018, two years before PIX launched: 0.5% of transaction value, averaged quarterly across a bank's total debit volume, with no single transaction above 0.8%.[16] Prepaid cards got a separate 0.7% cap in April 2023.[17] A study of the 2018 debit cap found merchant costs did fall after it took effect,[18] but that is due to the regulation.

Credit card fees are not capped, which makes them the one place a real PIX effect could plausibly show. The evidence here is thinner. Credit card fees fell from about 2.8% in 2014 to about 2.2% by 2021, driven by competition among payment processors and interchange caps introduced during that stretch.[19] Current sources cite 2%–5% for 2025–2026,[20] while PIX's own fee has stayed

[15] Agência Brasil. "Interest on Revolving Credit Cards Reaches 451.5% Per Year in Brazil." Agência Brasil, Sept. 2025.

[16] Castro, Daniel Tavares de, et al. "An Empirical Analysis of Debit Card Interchange Fee Regulation: Evidence from Brazil." Latin American Journal of Central Banking, vol. 4, no. 1, 2023.

[17] PYMNTS.com. "Brazil to Cap Prepaid Card Interchange Fees to 0.7% Starting April 2023." PYMNTS.com, Sept. 2022.

[18] Castro, Daniel Tavares de, et al. "An Empirical Analysis of Debit Card Interchange Fee Regulation: Evidence from Brazil." Latin American Journal of Central Banking, vol. 4, no. 1, 2023.

[19] Duarte, Angelo, et al. "Central Banks, the Monetary System and Public Payment Infrastructures: Lessons from Brazil's Pix." BIS Bulletin no. 52, Bank for International Settlements, Mar. 2022.

[20] "Credit Cards in Brazil 2026: Rates, the Rotativo Cap, IOF." *globecreditcards.com*, July 2026.

around 0.22%–0.33% the whole time.[21] Five-plus years in, credit card fees show no sign of moving toward PIX's price; if anything, the current range is as wide as, or wider than, where it stood in 2021.

Put together, both sides of the merchant-fee question line up with the interest-rate findings: wherever price actually moved, regulation, economic activity or processor competition moved it, not PIX. Cards and PIX look like two separate systems rather than substitutes that keep each other's prices in check. Card fees fund things PIX does not offer, including installment credit, chargeback protection, and loyalty programs, which may be exactly why merchants keep paying for card acceptance even as PIX stays cheaper.

VII. Policy Implications

If the actual goal is cheaper card credit, PIX does not look like the appropriate tool. The revolving rate is governed by a regulatory ceiling and by default risk, not competition between payment methods. The 2018 debit cap is the better precedent: direct regulation is what actually moved a comparable price before. If cheaper card credit is the goal, direct regulation is the most effective tool according to prior evidence.

VIII. Conclusion

Nothing here supports the idea that PIX's growth has lowered credit card interest rates, revolving or installment, or merchant fees in Brazil. Card interest rates rose through the exact years PIX usage grew fastest, and Brazil's Selic cycle explains that far better than PIX does. Merchant fees have stayed in roughly the same range as before PIX on the unregulated, credit side, and moved only because of a pre-PIX cap on the regulated, debit side. None of this contradicts PIX's real, well-documented effects elsewhere: deposit competition and account formation are genuine findings. It just means those effects stopped short of card pricing, which runs on its own logic of regulatory ceilings, risk-based pricing, and network economics, largely untouched by PIX.

[21] Duarte, Angelo, et al. "Central Banks, the Monetary System and Public Payment Infrastructures: Lessons from Brazil's Pix." BIS Bulletin no. 52, Bank for International Settlements, Mar. 2022.

IX. Limitations

Five annual points cannot support a regression or any claim of statistical significance; everything here is descriptive, and a strong correlation is not the same as proof of cause. The 2021 and 2022 PIX volume figures are estimates, not verified Central Bank data, which is a real gap. Every number in this paper is a national average, so there is no way to know whether individual banks behaved differently: for instance, whether one that gained heavily from PIX-driven deposits also happened to price its cards differently. The merchant-fee comparison rests on snapshots rather than a continuous series on the unregulated side, so it should be read as suggestive at most; the regulated side is better documented but governed by a rule that has nothing to do with PIX, which limits what it can say either way.

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